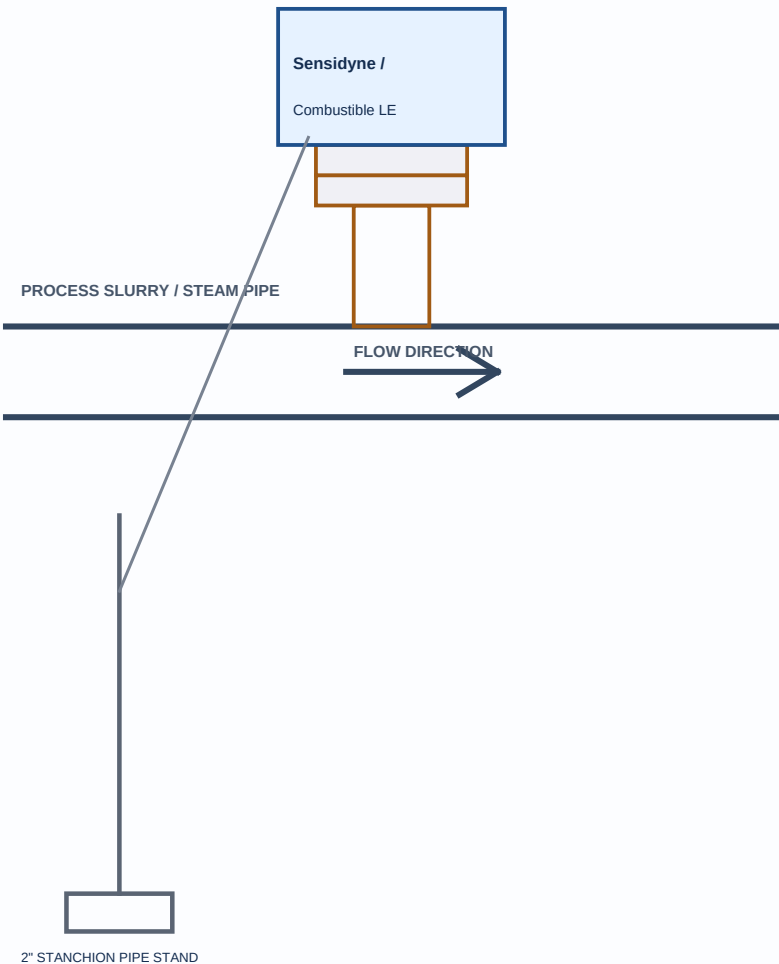


TEMPLATE TYP-02: COMBUSTIBLE LEL GAS ANALYZER

TYPICAL TAGS: AIT-60101 / AIT-60201

SERVICE: Explosion Hazard Lower Explosive Limit (LEL) Monitor | PLC I/O: 1756-IF16 (AI LEL) & 1756-IB32 (DI Alarm)

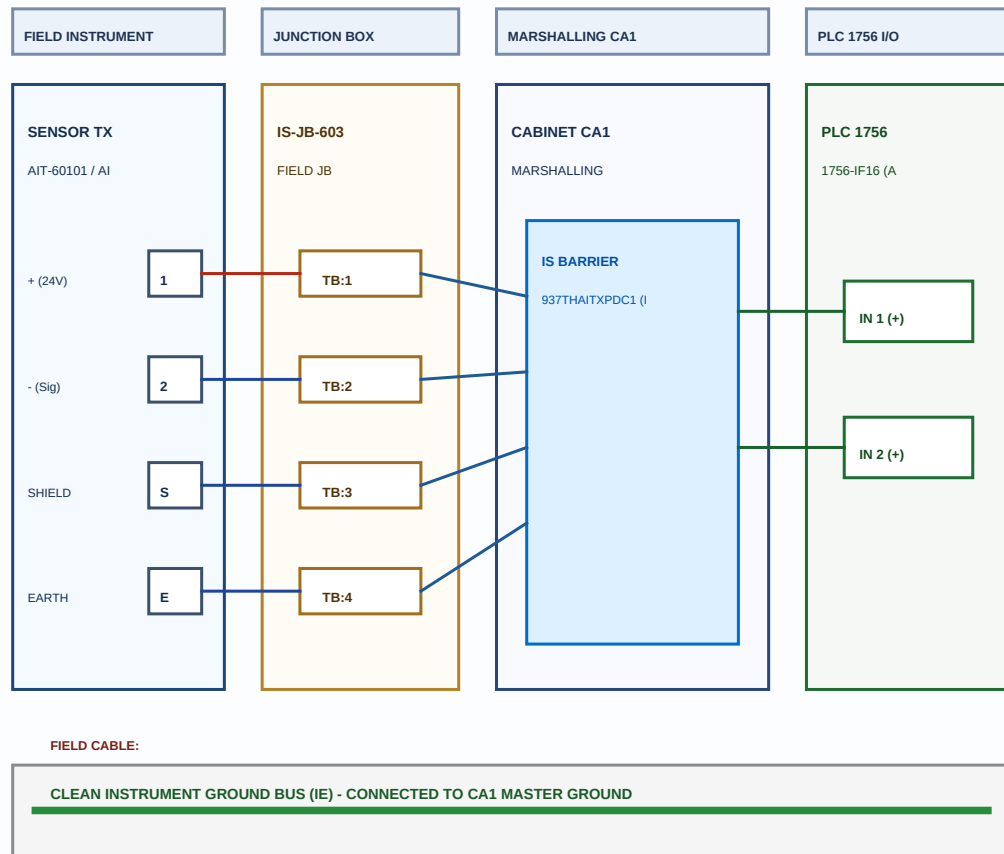
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	Ambient Air Diffusion / Spla..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66 Heavy-duty Ex..
Hazardous Class:	Class I, Div 1, Grp BCD / Ex..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	AI: 0-100% LEL (4-20mA) / DI: Gas Alarm
Power Supply:	24V DC (18-32V DC, 200mA max)
Field Cable Type:	3Cx1.5mm² Shielded Fire-Resistant (Bel..
Destination JB:	IS-JB-603 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THAITXPDC1 (IS Repeater Barrier)
PLC Module:	1756-IF16 (AI LEL) & 1756-IB32 (DI Ala..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: SENSIDYNE / SMARTGAS

MODEL: 925FGD Explosionproof Hydrocarbon Gas Detector

SERVICE: Explosion Hazard Lower Explosive Limit (LEL) Monitor

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Gas Detector Transmitter 925FG Sensidyne Infrared / Catalytic		1 EA
2	Weather Guard / Dust Protectio Hydrophobic SS316		1 EA
3	Calibration Adapter & Cup Molded PVC / Viton O-ring		1 EA
4	Explosionproof Cable Gland 3/4 SS316 Flameproof Ex d		1 EA
5	Wall / Stanchion Mounting Brac SS304 Universal Bracket		1 SET

ENGINEERING INSTALLATION NOTES

- Install 0.3m to 0.5m above floor for gases heavier than air, or near ceiling for lighter gases.
- Field sensor must undergo zero and span gas calibration prior to plant commissioning.
- Wiring must run in rigid galvanized steel conduit or approved armored flameproof cable.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONCOMBUSTIBLE LEL GAS DETECTOR HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-002
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
04 / 25